

Experimental Search for Time-Reversal Violation in Radiative Nuclear Beta Decay

Experimental Research Proposal

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Abstract

This proposal presents an experimental search for time-reversal violation in radiative nuclear beta decay. The study will investigate a T-odd momentum correlation among the beta particle, emitted photon, and recoil nucleus. The experiment will use a radioactive source or beam together with beta, gamma-ray, and recoil detection, precision timing, coincidence measurements, and event reconstruction. The aim is to determine whether a measurable asymmetry exists after accounting for electromagnetic final-state interactions and experimental systematic effects. The work is intended to be carried out within an established nuclear-physics collaboration at a suitable radioactive-beam laboratory.

1. Introduction

Time-reversal symmetry is a fundamental symmetry in physics. Weak nuclear interactions provide sensitive opportunities to test this symmetry and to search for effects beyond the Standard Model. Radiative beta decay is especially useful because, in addition to the beta particle and neutrino, a photon is emitted. Measuring the directions and momenta of the observable decay products can therefore provide a T-odd momentum correlation.

2. Research Problem

Experimental searches for time-reversal violation require extremely precise measurements because detector asymmetries, backgrounds, reconstruction errors, and electromagnetic final-state interactions can imitate a small T-odd signal. The central problem is to determine whether an observed asymmetry is consistent with known effects or indicates a genuine time-reversal-violating interaction.

3. Research Question

Can an experimental measurement of radiative nuclear beta decay detect a T-odd momentum asymmetry with sufficient precision to place a meaningful constraint on time-reversal violation?

4. Objectives

The main objective is to perform a precision experimental search for time-reversal violation in radiative nuclear beta decay. Specific objectives are to measure beta-particle, photon, and recoil observables; construct the T-odd momentum correlation; determine the corresponding asymmetry; calibrate the detector system; identify and reduce false asymmetries; evaluate systematic uncertainties; and compare the measured result with expected electromagnetic final-state

contributions.

5. T-Odd Momentum Correlation

A useful correlation is the scalar triple product $C_T = \mathbf{p}_{\beta} \cdot (\mathbf{p}_{\gamma} \times \mathbf{p}_{\text{recoil}})$, or an experimentally equivalent observable. Under time reversal, momenta reverse direction and the triple product changes sign. Events with positive and negative values can therefore be compared to determine a T-odd asymmetry.

6. Experimental Concept

A radioactive isotope will be delivered as a beam, implanted into a target, or held in an appropriate trapping system. The decay products will be detected in coincidence. Beta detectors will measure the beta particle, gamma-ray detectors will measure the emitted photon, and a recoil detector or recoil-tracking system will provide recoil information. A precision timing and data-acquisition system will associate the signals with individual decay events.

7. Experimental Apparatus

The proposed apparatus includes a radioactive-ion source or beam-delivery system, beta detectors, gamma-ray detectors, recoil detection, timing electronics, coincidence logic, magnetic or electrostatic fields where required, and a digital data-acquisition system. Detector geometry should be designed to provide good angular acceptance and to minimize instrumental asymmetry.

8. Experimental Procedure

The experiment will begin with detector installation and calibration. Energy, timing, position, efficiency, and threshold characteristics will be measured. During physics data taking, beta-gamma-recoil coincidence events will be recorded. For each event, the measured energies and directions will be reconstructed and used to calculate the T-odd correlation. Data will be collected under symmetry-related configurations when possible.

9. Calibration

Calibration will include energy calibration using appropriate reference sources, timing-offset measurements, detector-efficiency measurements, position and angular calibration, coincidence-window optimization, dead-time measurement, and background characterization. Detector performance will be monitored throughout data collection.

10. Systematic Uncertainties

Potential systematic effects include detector-efficiency differences, geometric acceptance, beta and gamma scattering, threshold effects, accidental coincidences, beam-position changes, magnetic-field imperfections, timing offsets, recoil reconstruction errors, and background events. Reversal of experimental configurations, detector symmetry, control runs, sideband studies, and independent analysis methods will be used to identify and quantify these effects.

11. Electromagnetic Final-State Interactions

Electromagnetic final-state interactions can generate a T-odd correlation even when the underlying interaction conserves time reversal. Therefore, a measured nonzero asymmetry cannot automatically be interpreted as fundamental time-reversal violation. The expected electromagnetic contribution must be calculated or constrained and included in the final interpretation.

12. Data Analysis

The primary observable will be an asymmetry between events with positive and negative values of the T-odd correlation. Statistical uncertainties will be obtained from the event sample. The analysis will include background subtraction where justified, stability checks, systematic-error evaluation, and comparison between independent data-taking configurations.

13. Possible Research Facilities

The experiment should be conducted through an established collaboration at a radioactive-beam laboratory. CERN-ISOLDE is a possible candidate because it provides radioactive ion beams and dedicated decay-study infrastructure. TRIUMF/TRINAT is particularly relevant because it has conducted experimental work on time-reversal violation in radiative beta decay. FRIB may also be considered depending on the required isotope, beam intensity, detector configuration, and collaboration opportunities.

14. CERN-ISOLDE Implementation Plan

If pursued at CERN-ISOLDE, the first stage would be a feasibility study with facility scientists and an appropriate collaboration. Isotope production yield, half-life, decay scheme, radiative branch, beta endpoint, photon energies, background conditions, detector acceptance, and available beam time would be evaluated. The project would then proceed through apparatus integration, commissioning, calibration, physics data taking, and final asymmetry analysis. The exact isotope and beamline should be selected in consultation with the host laboratory.

15. Student Contribution

The student researcher can contribute to detector preparation, calibration, timing and coincidence tests, supervised data acquisition, event reconstruction, asymmetry extraction, systematic-error studies, scientific documentation, and presentation of results. All work involving radioactive material must be conducted under the laboratory's radiation-safety procedures and within an authorized research collaboration.

16. Expected Results

The experiment may observe a statistically significant T-odd asymmetry or establish an improved limit on a time-reversal-violating contribution. Either outcome would provide useful information for precision tests of weak nuclear interactions. The project would also provide practical training in nuclear experimental physics, detector instrumentation, radioactive-beam techniques, timing electronics, and precision data analysis.

17. Feasibility and Limitations

The experiment is technically demanding and cannot realistically be performed as an independent school or university laboratory experiment without major infrastructure. It requires radioactive-beam access, specialized detectors, radiation-safety systems, experienced researchers, and substantial beam time. A realistic first stage is therefore participation in an established collaboration, beginning with detector development, calibration, analysis validation, or a pilot measurement.

18. Conclusion

This proposal develops a focused experimental approach to testing time-reversal symmetry in radiative nuclear beta decay. By measuring beta, photon, and recoil observables and carefully controlling instrumental and electromagnetic effects, the experiment can provide a sensitive test of a fundamental symmetry of weak interactions. The project is best pursued through an established international collaboration at a radioactive-beam facility.

References

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